3.9 | Exponential and Logarithmic Functions

AP CALCULUS AB | LESSON 2 | WARM UPS | RETRIEVAL BEFORE NEW EXAMPLES

Name: _____ Date: _____

1 | Multiple Choice NO CALCULATOR

Recent review | 3.9 Logarithmic Chain Rule

If $f(x) = x\ln(2x+3)$, which expression is $f'(x)$?

A. $\ln(2x+3) + \frac{x}{2x+3}$

B. $\frac{2}{2x+3}$

C. $\ln(2x+3) + \frac{2x}{2x+3}$

D. $2x\ln(2x+3)$

E. $\ln(2x+3) + 2x$

Show your reasoning here.

2 | Multiple Choice CALCULATOR PERMITTED

Application | Tangent Line to a New Function

For $f(x) = \ln(x^2+2) + e^{0.6x}$, which equation approximates the tangent line at $x = 1.2$?

A. $y - 1.93 = 3.29(x - 1.2)$

B. $y - 3.29 = 0.70(x - 1.2)$

C. $y - 3.29 = -1.93(x - 1.2)$

D. $y - 2.05 = 1.93(x - 1.2)$

E. $y - 3.29 = 1.93(x - 1.2)$

Show your reasoning here.

3.9 | Exponential and Logarithmic Functions

AP CALCULUS AB | LESSON 2 | LEARNING FOCUS & VOCABULARY | READ, HIGHLIGHT, CONNECT

Learning focus | I can...

- Differentiate exponential functions $e^{u(x)}$ and $a^{u(x)}$ using the Chain Rule.
- Differentiate natural-log and other logarithmic functions using $(\ln u)' = u'/u$.
- Use implicit differentiation to explain why logarithm and exponential rules work.
- Apply logarithm properties and logarithmic differentiation to products, quotients, and powers.

Vocabulary | Highlight the important words

Exponential function

A function where the variable appears in the exponent, such as e^x or 3^x.

Natural logarithm

The inverse of e^x, written $\ln x$; its domain is $x > 0$.

Base

The positive constant a in a^x; for logarithms, $a \neq 1$.

Logarithm

An exponent in another form: $\log_b a = c \iff b^c = a$.

Chain Rule

Multiply the outside derivative by the derivative of the inside function.

Change of base

Rewrite $\log_b x = \frac{\ln x}{\ln b}$ to differentiate.

Logarithm properties

Product, quotient, and power rules that rewrite products as sums and powers as factors.

Logarithmic differentiation

Take natural logs, simplify, differentiate implicitly, and solve for the derivative.

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AP CALCULUS AB | LESSON 2 | MEANING & METHOD | REFERENCE NOTES: REPEAT BOTH LESSONS

1 | Exponential functions and the Chain Rule

$\frac{d}{dx}e^x = e^x \quad \frac{d}{dx}e^{u(x)} = e^{u(x)}u'(x) \quad \frac{d}{dx}a^{u(x)} = a^{u(x)}\ln(a)\,u'(x)$

2 | Natural logarithm and other bases

$\frac{d}{dx}\ln x = \frac{1}{x} \quad \frac{d}{dx}\ln(u) = \frac{u'}{u} \quad \frac{d}{dx}\log_b x = \frac{1}{x\ln b}$

Use the Chain Rule for an inside expression. For $\ln(u)$, the logarithm requires $u > 0$.

3 | Properties of logarithms

$\log_b(MN) = \log_b M + \log_b N \quad \log_b(M/N) = \log_b M - \log_b N \quad \log_b(M^k) = k\log_b M \quad \log_b a = \frac{\ln a}{\ln b}$

For these identities, the logarithm arguments must be positive. The base must satisfy $b > 0,\ b \neq 1$.

4 | Logarithmic differentiation method

1. Begin with $y = \text{product} \,/\, \text{quotient} \,/\, \text{power}$. Take $\ln$ of both sides.
2. Use log properties to bring down exponents and split products/quotients.
3. Differentiate implicitly: the left side becomes $\frac{y'}{y}$.
4. Multiply by y and substitute the original expression back.

Notes / questions:

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AP CALCULUS AB | LESSON 2 | EXAMPLES & YOU TRY ITS | ORIGINAL EXAMPLE 7 AND A MATCHED PRACTICE

Example 7 | Change of base and a radical

From Section 3.9 source notes

Use logarithm properties to rewrite, then differentiate $y = \log_5 \sqrt{x}$.

Show work and justify each step.

You Try It 7 | Change of base and a radical

Matching practice (new numbers)

Use logarithm properties to rewrite, then differentiate $y = \log_3 \sqrt{x}$.

Show work and justify each step.

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AP CALCULUS AB | LESSON 2 | EXAMPLES & YOU TRY ITS | ORIGINAL EXAMPLE 8 AND A MATCHED PRACTICE

Example 8 | Logarithmic differentiation

From Section 3.9 source notes

Use logarithmic differentiation to find $\frac{dy}{dx}$ for $y = \frac{x\sqrt{x^2+1}}{(x+1)^{2/3}}$.

Show work and justify each step.

You Try It 8 | Logarithmic differentiation

Matching practice (new numbers)

Use logarithmic differentiation to find $\frac{dy}{dx}$ for $y = \frac{x\sqrt{x^2+4}}{(x+2)^{2/3}}$.

Show work and justify each step.

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AP CALCULUS AB | LESSON 2 | EXAMPLES & YOU TRY ITS | ORIGINAL EXAMPLE 9 AND A MATCHED PRACTICE

Example 9 | Differentiate a constant-base exponential

From Section 3.9 source notes

Find $\frac{dy}{dx}$ if $y = 2^x$. Use logarithmic differentiation to justify the rule.

Show work and justify each step.

You Try It 9 | Differentiate a constant-base exponential

Matching practice (new numbers)

Find $\frac{dy}{dx}$ if $y = 5^x$. Use logarithmic differentiation.

Show work and justify each step.

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AP CALCULUS AB | LESSON 2 | EXAMPLES & YOU TRY ITS | ORIGINAL EXAMPLE 10 AND A MATCHED PRACTICE

Example 10 | Apply the general exponential rule

From Section 3.9 source notes

Find $\frac{dy}{dx}$ if $y = 3^x$.

Show work and justify each step.

You Try It 10 | Apply the general exponential rule

Matching practice (new numbers)

Find $\frac{dy}{dx}$ if $y = 4^x$.

Show work and justify each step.

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Example 11 | Combine all three families

From Section 3.9 source notes

Find the derivative of $g(x) = e^{5x} + 7^x + \ln(x^3 + 4)$.

Show work and justify each step.

You Try It 11 | Combine all three families

Matching practice (new numbers)

Find the derivative of $h(x) = e^{3x} + 5^x + \ln(x^2 + 9)$.

Show work and justify each step.